

✓ Task 2 (Edge Detection)

Apply an edge detection algorithm to detect the road edges in the image road.jpg. Fine-tune the thresholds to exclusively detect the road edges while minimizing detection of other features in the image. Use gray colormap to display.

```
import cv2 as cv
from matplotlib import pyplot as plt

image = cv.imread('/content/road.jpg', cv.IMREAD_GRAYSCALE)

plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")
plt.show()
```



Original Image



```
edges = cv.Canny(image, 400, 900)
plt.imshow(edges, cmap='gray')
plt.title("Original Edge Detection")
plt.axis("off")
plt.show()
```



Original Edge Detection



```
# Reduce noise (if any)
img_blur = cv.GaussianBlur(image, ksize = (3, 3), sigmaX = 0, sigmaY = 0)

# Low and high Thresholds
t = 190
T = 804

edges = cv.Canny(img_blur, t, T)
plt.imshow(edges, cmap='gray')
plt.title("Road Edge Detection")
plt.axis("off")
plt.show()
```



Road Edge Detection

